

From Spin-Selective Protein Chemistry to Conscious Neural Rendering

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Abstract

The same neural input can have different consequences because a receiving cell is a stateful physical system. Receptors, proteins, redox condition, recent activity, inhibition, metabolism, and molecular configuration all help determine what an arriving signal can do. Spin chemistry supplies one candidate source of this state dependence: electron or nuclear spin dynamics can change reaction branching and leave a classical product, conformation, or signaling state. Recent experiments have made the measurement problem more tractable by demonstrating spin readout in genetically encoded proteins, room-temperature optically detected magnetic resonance in living cells, and engineered control of spin-correlated radical-pair dynamics in a transgenic animal. These results establish measurement and actuation platforms; the endogenous neural mechanism remains to be found. This paper defines the causal ladder needed to find or reject it: identified spin system, preparation, lifetime, spin-selective yield, protein or receptor consequence, cellular consequence, network consequence, and task-specific perceptual, mnemonic, choice, action, or report consequence. Three nested hypotheses distinguish transient spin-conditioned classical state, memory participation, and literal spin-stored task information. Twenty numbered equations formalize spin evolution, reaction yield, protein state, receiver transformation, receiver-relative neural difference, causal mediation, held-out information, chain strength, a spin-specific interaction, temporal ordering, downstream-state rescue, nested predictive models, and a conjunctive decision gate. The neural stage is then connected to Self Aware Networks, Neural Array Projection Oscillation Tomography, and Phase Wave Differentials. The experimental program uses isotope and field controls, reaction-yield-detected magnetic resonance, matched non-spin chemistry, mediator blockade, downstream-state rescue, and content-specific neural tests. A positive result would identify a molecular state-setting route into conscious neural rendering. A null result would localize which version of the mechanism fails. Keywords: spin chemistry; radical pairs; protein state; neural rendering; receiver-state dependence; Phase Wave Differential; NAPOT; consciousness; causal mediation; matched rescue

1. The problem: identical input does not meet an identical receiver

Imagine that the same presynaptic pattern reaches the same dendritic branch twice. On the first trial, a receptor complex occupies one conformation, local redox chemistry has one value, inhibition is weak, and the cell is metabolically ready. On the second, recent activity has changed receptor availability, phosphorylation, membrane conductance, calcium, and protein state. Nearly identical spikes can therefore produce different output.

Neuroscience already explains much of this state dependence through ion channels, receptors, neuromodulators, membrane voltage, calcium, inhibition, plasticity, and metabolism. The unresolved question is more specific: can a spin-selective molecular event become one of the state variables that changes what a neural receiver does next?

A quantum effect located somewhere in a neuron is not enough. A mechanism must identify:

1. the spin-bearing species and its biological location;
2. preparation and readout;
3. relevant lifetime at the measured temperature and environment;

4. a spin-selective product or conformation;
5. a protein, receptor, transcriptional, or plasticity consequence;
6. a membrane, synaptic, or cellular consequence;
7. a circuit timing, phase, routing, inhibition, or persistence consequence;
8. a task-specific memory, percept, choice, action, or report consequence;
9. an intervention that breaks the predicted link; and
10. a matched rescue that restores the downstream state.

This molecule-to-report ladder is the paper's central contribution. It converts a broad quantum-consciousness intuition into separable measurements.

2. Established physical and biological foundations

2.1 Protein state already changes neural transformation

Ligand binding, phosphorylation, oxidation, reduction, mechanical stress, local fields, temperature, pH, and molecular interactions change protein conformation and kinetics. In neurons, these changes alter receptor opening, ion-channel probability, transmitter release, trafficking, gene expression, and plasticity. The spin hypothesis adds a discriminator to these established routes: changing spin dynamics under a matched chemical condition must shift a downstream state in the direction predicted by a spin-selective reaction model.

2.2 Spin chemistry can select a chemical branch

A radical pair can be born in a singlet or triplet configuration. Hyperfine interactions, Zeeman terms, exchange, dipolar coupling, relaxation, and reaction rates determine singlet-triplet evolution and product yield. Flavin, cryptochrome, and magnetoreception studies establish both the mechanism and its constraints [1-3,18-20]. Effects depend on molecule, geometry, field, lifetime, temperature, and environment.

The short spin event need not persist for a thought. Its reaction products or protein conformations can become classical states that outlast the spin dynamics. This biochemical amplification route is the least demanding bridge from quantum chemistry to neural state.

2.3 Protein-spin measurement has advanced from purified systems to living organisms

Feder et al. realized an optically addressable triplet spin qubit in enhanced yellow fluorescent protein [4]. Coherent microwave control and a 16 ± 2 microsecond coherence time under decoupling were demonstrated at liquid-nitrogen temperature. The protein was expressed in mammalian cells, and room-temperature optically detected magnetic resonance reached up to 8 percent contrast in bacterial cells. This establishes a genetically encodable spin platform, not a native cognitive mechanism.

Abrahams et al. engineered MagLOV proteins whose flavin-associated spin-correlated radical-pair dynamics produce room-temperature optically detected magnetic resonance in living bacterial cells [22]. They demonstrated single-cell detection, microenvironment sensitivity, and spatial localization. Burd et al. then altered red-fluorescent-protein/flavin radical-pair dynamics with static and radiofrequency fields in vitro and in transgenic *C. elegans*, reporting a coherence time greater than 4 nanoseconds [23]. These are engineered, non-native biochemical systems. They prove that the required measurements and interventions are becoming experimentally available.

3. Earlier proposals and the residual mechanism

Hu and Wu proposed in 2004 that nuclear spins in neural membranes and proteins form consciousness and memory substrates influenced by action potentials and paramagnetic molecules [5]. Fisher proposed in 2015 that phosphorus spins protected in Posner molecules participate in quantum processing, calcium release, neurotransmission, and firing [6]. Hameroff and Penrose developed a microtubule-centered quantum-collapse architecture [7]. These programs own genuine earlier quantum- neural constituents.

Nair, Zadeh-Haghighi, and Simon directly modeled NMDAR magnetic-field effects in 2024 using candidate serine- or tyrosine-based oxyradical/magnesium radical pairs [21]. Their calculations predict field- and isotope-dependent singlet yields that could influence phosphorylation, receptor expression or activity, and calcium. The nominated magnesium radical has not been observed in a biological system, its reaction and relaxation rates were not measured, and parts of the magnetic- isotope literature remain contested. The paper therefore supplies a direct NMDAR radical-pair model and a precise experimental target, not a demonstrated receptor spin transducer.

Cavelier's 2026 paper proposes that conscious sensory and emotional experience is embedded in NMDAR-based radical pairs and that their spin states transfer information into a proposed quantum- iEEG wavefunction biconditionally related to spike frequency and timing [8]. His earlier verified work had already modeled ion-channel electrochemistry (1996), reduced-flavin quantum chemistry (2001), and FAD-enzyme protein fluctuations (2006) [24-26]. Those are genuine earlier molecular constituents. The presently recovered earlier record does not join them to radical-pair spin, NMDAR-based conscious content, spike coding, or distributed neural rendering.

The residual problem is therefore not "quantum spin in the brain" in general. It is the joined operation:

spin-sensitive molecular state -> protein/receptor transformation -> receiver-relative neural difference -> distributed rendering -> content-specific memory, choice, action, or report.

4. Development of the present proposal

Stage	Source-bound contribution	Date boundary
August 2011	self as a prediction pattern in the neocortex, whole-neocortex observer, and internally constructed world-model ancestry	owner-metadata-supported recording; current public page exposes a 2025 date; no spin atom assigned [27]
August 2012	recursive EEG-to-light/sound feedback, coordinated neural-wave language, and mind-as-projector world-model architecture	live platform-dated public sources; no spin atom assigned [28,29]
2017	spatial neural computation, changing 3D/4D patterns, feedback/broadcast distinction, and ATP information-energy bridge	public Neural Lace Podcast, 11 April 2017 [9]
June 2022	radical-pair spin selection, field-window questions, flavins, ETF, mitochondrial ATP, receptor state, tonic/phasic activity, and learning	immutable public Git [10-13]
December 2023	protein spin, protein-spin memory, weak-field effects, receptors, flavins, mitochondria, and NAPOT 7	private Micah-authored prompt; assistant text separately attributed [14]
January 2024	mature protein-spin branch joined publicly to PWD, neural rendering, memory, and consciousness	immutable public Git [15]
2024	broader molecular-to-oscillatory, plasticity, PWD, and rendering architecture	published book [16]
February 2025	protein-spin-specific signaling, spin-dependent protein memory, PWD/global spectral mapping, and ATP reinforcement are restated in the public OCA draft	immutable public Git, 21 February 2025 [30]
2026	equations, measurement ladder, causal experiments, and comparison synthesis	current work; not backdated

The mature public protein-spin formulation is fixed in January 2024. The June 2022 sources remain earlier, narrower ancestors and retain their own dates. The 2011 source is retained as authorship genealogy rather than strict public-priority proof because its present watch page does not expose the owner-recorded 2011 date. The two 2012 sources have live platform timestamps and establish the recursive-feedback and projector architecture, but not spin chemistry. The February 2025 Git source confirms public continuity of the protein-spin/PWD architecture; it does not move the January 2024 public start date or supply the experiments developed here.

5. The proposed operation, before its name

Light, ligand binding, redox state, isotope composition, or a field changes the spin evolution of a specific molecular pair. That shift slightly favors one reaction product or protein conformation. The resulting classical state changes receptor current, channel conductance, vesicle release, transcription, or plasticity. The receiving neuron consequently transforms the same input differently. Across a connected circuit, changed transformations alter timing, propagation, inhibition, phase relations, or persistence. Those differences can alter what is recognized, remembered, selected, rendered, acted upon, or reported.

Self Aware Networks (SAN) names the wider receive-transform-project-feedback architecture. Neural Array Projection Oscillation Tomography (NAPOT) names the proposed multiscale reconstruction and re-expression of distributed neural relations. A Phase Wave Differential (PWD) is the receiver- relative, context-dependent difference between an event and the relation expected along that route. A spin-conditioned protein state is one possible upstream variable that changes a PWD. It is not the viewer; the recurrent network performs the observing operation by being changed and using the change.

This is why PWD is distinct from phase coding and traveling waves. Phase coding describes information associated with phase; traveling-wave language describes propagation. PWD specifies a receiver, an expected relation, a measured multivariate deviation, and a consequential downstream change. Phase and propagation can be components of that deviation without exhausting it.

6. Three nested hypotheses

Hypothesis A: spin chemistry sets a classical protein state

A short spin-selective reaction changes product yield or conformation. The classical state lasts long enough to alter a neural variable. No long-lived neural qubit is required.

Hypothesis B: spin-conditioned state participates in memory

The classical state changes plasticity, protein synthesis, receptor configuration, encoding, or retrieval in a task-specific manner. Spin chemistry participates in a memory operation without storing semantic content in the spin itself.

Hypothesis C: spin itself stores task-relevant information

A prepared spin state persists, is selectively written and read, and predicts memory content after ordinary chemical states are controlled. This strongest formulation requires lifetime, addressability, capacity, selective readout, and causal evidence. A failure of C leaves A and B open; evidence for A cannot be reported as evidence for C.

7. Mathematical framework

7.1 Spin evolution and yield

For an identified spin system with density operator $\rho_s(t)$, define

$$H_s(t) = H_Z(t) + H_{hf} + H_{ex} + H_{dip}. \quad (1)$$

Open-system evolution is

$$\begin{aligned} & \frac{d\rho_s}{dt} \\ &= -\frac{i}{\hbar} [H_s(t), \rho_s(t)] + L_{relax}[\rho_s(t)] \\ & \quad - \frac{1}{2} \{K, \rho_s(t)\}, \end{aligned} \quad (2)$$

where $K = k_S P_S + k_T P_T$. Integrated yields are

$$\begin{aligned} \Phi_S &= k_S \int_0^\infty Tr[P_S \rho_s(t)] dt, \\ \Phi_T &= k_T \int_0^\infty Tr[P_T \rho_s(t)] dt. \end{aligned} \quad (3)$$

For isotope, field, light, orientation, or mutation intervention u ,

$$\Delta \Phi(u) = \Phi_S(u) - \Phi_S(u_0). \quad (4)$$

The reference u_0 must match mass, temperature, toxicity, illumination, redox chemistry, and protein abundance as closely as the intervention permits.

7.2 Yield to protein state

Let $z(t)$ include conformation, binding, phosphorylation, oxidation, localization, or channel state:

$$\frac{dz}{dt} = f_z(z, c, t) + G_z \Phi(t) + \eta_z(t). \quad (5)$$

Hypothesis A requires a nonzero G_z under biologically plausible conditions and successful matched-spin controls.

7.3 Receiver transformation

For receiver i ,

$$R_i(t) = [g_i(t), V_i(t), I_i(t), p_i(t), m_i(t), z_i(t), h_i(t)], \quad (6)$$

where the components describe conductance/receptor state, membrane voltage, inhibition, transmission probability, metabolic readiness, protein state, and recent history. For input X_{ji} ,

$$Y_i(t+\delta) = F_i(X_{ji}(t), R_i(t), C_i(t)) + \varepsilon_i(t). \quad (7)$$

7.4 Expected relation and PWD

The tonic or expected output is

$$\mu_{ji}(t) = E[Y_i(t+\delta) | X_{ji}(t), R_i(t), C_i(t), H_i(t)]. \quad (8)$$

The signed receiver-relative difference is

$$D_{ji}(t) = Y_i(t+\delta) - \mu_{ji}(t). \quad (9)$$

For conditional covariance $\Sigma_{ji}(t)$, its standardized magnitude is

$$\Pi_{ji}(t) = \sqrt{D_{ji}(t)^T \Sigma_{ji}(t)^{-1} D_{ji}(t)}. \quad (10)$$

The signed vector remains primary because it preserves which timing, phase, amplitude, duration, route, or molecular component changed.

7.5 Causal mediation

For spin intervention U, protein state Z, neural outcome N, and behavior/report B,

$$U \rightarrow Z \rightarrow N \rightarrow B. \quad (11)$$

A natural indirect-effect contrast through Z is

$$NIE_Z = E[B(U=1, Z_0) - B(U=1, Z_0)]. \quad (12)$$

This estimand requires declared causal-identification assumptions and cannot be inferred from correlation alone.

7.6 Incremental information and chain strength

For spin observables Q and conventional covariates C,

$$I(N; Q | C) > 0 \quad (13)$$

is evaluated with a preregistered, bias-corrected estimator, a surrogate or permutation null, uncertainty intervals, and held-out replication. Exact positivity is not inferred from a noisy point estimate. Held-out model improvement is

$$\Delta L = L_{test}(N | C, Q) - L_{test}(N | C). \quad (14)$$

For standardized, directionally coded link estimates, define the diagnostic composite

$$\Gamma_{chain} = \gamma_{U \rightarrow Z} \gamma_{Z \rightarrow N} \gamma_{N \rightarrow B}, \quad \Pr(|\Gamma_{chain}| > \gamma_{min}) \geq 1 - \alpha. \quad (15)$$

Every link remains separately reported. The coefficients in Equation (15) must be standardized, directionally coded, estimated under the declared model, and accompanied by their joint uncertainty. The

product is not a posterior probability and cannot turn a weak or missing link into evidence. Equation (20), rather than the magnitude of Equation (15), is the full-chain decision gate. A large molecular effect cannot conceal a null neural or behavioral link.

7.7 Identifiability, temporal order, rescue, and model comparison

A spin-active intervention can also change temperature, light exposure, protein abundance, folding, localization, redox chemistry, and ordinary reaction kinetics. Spin specificity is therefore an interaction, not a raw difference. Let $S=1$ denote the nominated spin-active molecule and $S=0$ a matched spin-insensitive variant; let $U=1$ denote the field, isotope, radiofrequency, orientation, or preparation contrast. For primary endpoint Y ,

$$\begin{aligned} \Delta_{spin} &= (E[Y|S=1, U=1] - E[Y|S=1, U=0]) \\ &\quad - (E[Y|S=0, U=1] - E[Y|S=0, U=0]). \end{aligned} \quad (16)$$

The direction and minimum meaningful magnitude of Δ_{spin} must be predicted from the declared spin-chemistry model before the confirmatory data are inspected. The interaction is estimated separately at the product/protein, neural, and behavior/report levels. A matched variant must be characterized for abundance, folding, localization, binding, illumination response, temperature, redox state, and ordinary kinetics. Where feasible, the conclusion must reproduce across two orthogonal spin contrasts rather than depend on one mutant.

Let τ_Z , τ_N , and τ_B be onset times for the protein-state, neural, and behavior/report consequences. Temporal support requires

$$\begin{aligned} \Theta_{order} &= \Pr(\tau_Z < \tau_N < \tau_B | D) \\ &\geq 1 - \alpha_{time}. \end{aligned} \quad (17)$$

The time-resolution limit, onset estimator, uncertainty framework, and allowable ties are preregistered. The probability can be computed under a declared Bayesian, bootstrap, or simulation-based measurement-error model. If the sensors cannot resolve the ordering, the result is indeterminate rather than positive. Equation (17) does not by itself establish causation; it rules out versions in which the proposed mediator reliably occurs after its alleged consequence.

The two-rescue design distinguishes the nominated spin actuator from its downstream classical transducer. Let z_1 be the protein/receptor state produced by the active spin route and z_0 its control state. Under spin-path blockade $U=0$, define

$$\begin{aligned} \Delta_Z^{rescue} &= E[N | do(U=0, Z=z_1)] \\ &\quad - E[N | do(U=0, Z=z_0)], \\ |\Delta_Z^{rescue} - \Delta_U^N| &\leq \varepsilon_R, \end{aligned} \quad (18)$$

where $\Delta_U^N = E[N | do(U=1)] - E[N | do(U=0)]$, and ε_R is a preregistered equivalence margin. This form remains defined when the active-route contrast is zero: a zero contrast simply means there is no neural effect to rescue. A successful downstream rescue requires matched direction and equivalence, while a rescued neural state with no rescued content, choice, action, or report supports only the neural portion of the chain.

Three nested held-out models prevent ordinary biochemical state from being mistaken for spin-specific information:

$$\begin{aligned}
S_A &= CVScore(N|C), \\
S_B &= CVScore(N|C,Z), \\
S_C &= CVScore(N|C,Z,Q).
\end{aligned}
\tag{19}$$

Model B tests the classical protein-state bridge. Model C tests whether measured spin observables Q add value after conventional covariates C and protein state Z are included. The score is defined so that larger is better. Its scoring rule, folds, grouping unit, leakage controls, and thresholds δ_Z and δ_Q are frozen in advance. Hypothesis A can succeed when $S_B - S_A > \delta_Z$ even if $S_C - S_B \leq \delta_Q$; Hypothesis C requires the latter increment plus selective write/read evidence.

No weighted average may allow one strong link to hide a missing one. For a preregistered set of binary gates g_k , define

$$\begin{aligned}
G_{spin} &= g_{specificity}, g_{order}, \\
&g_{U \rightarrow Z}, g_{Z \rightarrow N}, g_{N \rightarrow B}, g_{blockade}, \\
&g_{rescue}, g_{replication}.
\end{aligned}
\tag{20}$$

A complete spin-to-rendering result requires $G_{spin}=1$. This is a transparent conjunctive decision rule, not a probability that the theory is true. Molecular, cellular, neural, and behavioral results remain interpretable at their achieved level when the full gate is zero.

8. Candidate systems ranked by evidentiary role

Candidate	Established value	Missing link
cryptochrome/flavin radical pair	mature spin-chemistry lineage	neuronal expression, preparation, biochemical gain, neural consequence
red fluorescent protein/flavin and MagLOV	room-temperature living-cell and in-vivo engineered measurement/actuation [22,23]	native neural function; engineered reporters may perturb the system
enhanced yellow fluorescent protein spin qubit	genetic encodability and coherent-control/readout platform [4]	biologically relevant temperature, sensitivity, and endogenous target coupling
mitochondrial ETF or related flavoprotein	source-genealogy bridge to redox and ATP	reproducible spin-selective response under neural conditions, separated from workload
NMDAR-associated oxyradical/magnesium pair	direct 2024 theoretical receptor route [21]	biological observation of the pair, measured rates, receptor-current mediation
phosphorus/Posner system	earlier nuclear-spin processing model [6]	independent biochemical feasibility and relationship to protein-state route

Candidates are ranked by biological location, lifetime, temperature, field scale, reaction yield, readout sensitivity, perturbability, downstream gain, and task linkage - not by metaphor.

9. Experimental program

9.1 Purified molecular decision

Reconstruct the candidate pair or protein. Measure spin observables, reaction products, conformation, binding, and kinetics across field, orientation, radiofrequency, light, ligand, redox, temperature, and timing. Reaction-yield-detected or optically detected magnetic resonance can test the nominated spin transition. Isotope and residue substitutions must include matched non-spin controls. Candidate variants must be characterized for abundance, folding, localization, absorption, binding, redox chemistry, and ordinary reaction kinetics. Whenever feasible, reproduce the predicted effect with two orthogonal spin contrasts. The model must predict the direction and kinetic window before confirmatory measurement.

9.2 Cellular transducer decision

Express or preserve the candidate in a neuron, organoid, or relevant cell model. Record protein state, receptor current, voltage, calcium, release, ATP/redox, temperature, and viability. A valid result is a replicated spin-sensitive protein or receptor change that precedes and mediates a specific cellular effect. Engineered fluorescent systems should first be used as calibrated sensors and positive controls, then separated from any claim about the native mechanism.

9.3 Receiver-relative network decision

Repeat a defined input while manipulating the candidate state. Fit Equations (7)-(10) on training data and predict held-out timing, phase, inhibition, release, splay, propagation, or another declared output. Compare against receptor, voltage, calcium, redox, metabolism, firing-rate, and arousal models.

9.4 Blockade and two rescues

Block the mediator while preserving general physiology. The predicted neural effect should vanish. Then test:

- spin-path rescue: restore the spin-sensitive condition;
- downstream-state rescue: impose the measured classical protein/receptor state without restoring the spin manipulation.

If downstream-state rescue succeeds while spin-path rescue fails, the classical transducer remains relevant and the nominated spin actuator is weakened.

9.5 Content, memory stage, action, and report

Predefine circuit, task epoch, content variable, and report or action. Separate memory encoding, consolidation, retrieval, reconsolidation, and report. A content effect must survive controls for global impairment, sedation, stress, movement, difficulty, and arousal.

9.6 Confirmatory preregistration matrix

The confirmatory program freezes one row for each causal level before data collection:

Level	Experimental unit	Primary endpoint	Confirmatory contrast	Required negative control	Pass condition
spin system	purified molecule or identified protein complex	singlet/triplet yield, ODMR/RYDMR response, or declared spin observable	active versus control U within S=1	matched S=0, temperature, light, mass, redox, abundance, folding, localization, ordinary kinetics	predicted Δ_{spin} direction and minimum effect
protein transducer	molecule, cell, or compartment	product, conformation, binding, phosphorylation, current, or trafficking	intervention-by-spin-active interaction	non-spin chemical perturbation producing matched bulk change	mediator changes after spin event and before neural outcome

Level	Experimental unit	Primary endpoint	Confirmatory contrast	Required negative control	Pass condition
neural receiver	identified cell, branch, organoid, or circuit	declared component of $D_{(ji)}$, $\Pi_{(ji)}$, release, inhibition, timing, or propagation	active route versus matched route	firing-, calcium-, ATP-, redox-, and arousal-matched alternatives	held-out increment, blockade, and downstream-state rescue
memory/content	preregistered task and stage	encoding, consolidation, retrieval, percept, choice, or action variable	task-specific contrast	difficulty, movement, stress, sedation, and report controls	mediated content effect without global impairment
replication	independent batch, laboratory, or held-out cohort	same primary contrast	frozen model and threshold	full null reporting	directionally consistent estimate and corrected uncertainty

Randomization, blinding, exclusions, sample-size rationale, estimator, equivalence bounds, missing-data handling, multiplicity correction, and stopping rule are frozen before confirmatory acquisition. Exploratory candidate discovery remains labeled exploratory and cannot be silently pooled with the confirmatory test.

10. Alternative explanations and controls

Every positive result is tested against heating, vibration, light intensity, phototoxicity, isotope mass effects, nonspecific ROS/redox change, expression artifacts, ligand and pH differences, membrane damage, toxicity, ATP or calcium shifts without spin specificity, movement, stress, instrument cross-talk, analysis flexibility, and multiple testing. The confirmatory design preregisters the candidate, contrast, mediator, neural endpoint, behavioral endpoint, exclusions, and minimum meaningful effect.

11. Longitudinal comparison with Cavelier

The comparison contains three nonexclusive truths.

1. Cavelier's 1996-2006 work is earlier on ion-channel electrochemistry, reduced-flavin quantum chemistry, and protein/cofactor fluctuations [24-26].
2. Micah's public January 2024 source is earlier on the joined protein-spin $\rightarrow$ receiver-relative PWD $\rightarrow$ distributed neural rendering, memory, and consciousness architecture [15].
3. Nair et al. are earlier than Cavelier on the direct published NMDAR radical-pair model [21].

Cavelier's 2026 formulation is functionally close because it joins radical pairs, NMDAR, spin-state information, spike timing/frequency, and conscious experience [8]. Under the operational and narrative lenses this is a partial-match candidate to the mature Micah chain. Under the detective and longitudinal lenses, the verified earlier Cavelier corpus supplies molecular constituents but no recovered conscious spin-to-spike join; that joined subject appears later. The record currently establishes chronology and functional overlap. It contains no evidence sufficient to infer access, dependent derivation, or intent.

12. Falsification and interpretation

The corresponding hypothesis is narrowed or rejected if:

1. no identified system shows the predicted spin-sensitive yield under plausible neural conditions;
2. matched thermal, optical, isotope, or chemical controls remove the effect;
3. a molecular effect produces no protein or receptor change;
4. the protein state adds no causal or predictive neural value;
5. toxicity, heating, redox, calcium, metabolism, or global firing fully explains the neural result;
6. blockade does not remove the effect or matched rescue fails;

7. behavior/report changes bypass the proposed molecular and neural mediators;
8. PWD variables add no out-of-sample value;
9. Hypothesis C fails lifetime, selective write/read, capacity, or content tests.

The strongest supported conclusion must match the highest completed link. Molecular spin control is a molecular result. A mediated change in receptor current is a cellular result. A replicated change in a content-bearing distributed pattern and report through the full chain is evidence for a molecular contribution to conscious neural rendering.

13. Conclusion

The decisive question is whether a named spin-sensitive molecular event creates a causal, task-relevant difference in what a neural receiver does. The tools now exist to prepare, perturb, and read spin-correlated protein chemistry in living systems. The missing experiment is the joined neural chain.

This framework begins with measured spin dynamics, requires a classical biochemical transducer, tests receiver-relative neural change, and ends with controlled memory, perception, choice, action, or report. Within SAN and NAPOT, a supported protein-spin route would explain one way receiver state changes a PWD and thereby changes a distributed rendering. The recurrent brain-body network remains the system that receives, transforms, projects, tests, and reuses the consequence.

Author contributions and AI-assisted development

Micah Blumberg conceived the theory, defined the mechanism and research questions, supplied the historical source corpus, and is the manuscript author. AI tools assisted with source routing, mathematical formalization, drafting, and internal consistency checks. Blumberg reviewed the scientific claims and retains responsibility for the manuscript. Historical sources, external evidence, current synthesis, and proposed experiments are identified separately.

Data and materials availability

This theoretical and methods paper reports no new participant, animal, or wet-laboratory data. Its governed source manifest, claim-source genealogy atlas, validation receipts, and figure specifications accompany the manuscript package. External sources are identified by DOI or stable public URL. Private development sources are used only for authorship genealogy and are not presented as public priority evidence.

Ethics statement

The manuscript proposes future molecular, cellular, neural, and behavioral experiments. Any study involving animals or human participants will require prospective institutional review, welfare protections, consent where applicable, and a preregistered protocol. No such experiment is reported here.

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